

TECHNICAL REPORT

ESSE

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TECHNICAL REPORT

Renewable Energy Adoption Barrier Advisor

Environmental Problem: Low Adoption of Renewable Energy

Relevant SDG: SDG 7 – Affordable and Clean Energy

1. Abstract

The increasing demand for energy and continued dependence on conventional energy sources have created significant environmental challenges, including greenhouse gas emissions, air pollution and climate change. Renewable energy sources such as solar, wind and hydropower provide cleaner alternatives to conventional energy. However, the availability of renewable-energy technologies does not necessarily result in their widespread adoption. Many potential users hesitate to adopt renewable energy because of financial concerns, lack of awareness, uncertainty about maintenance, reliability concerns and difficulty understanding available government support.

This report proposes a **Renewable Energy Adoption Barrier Advisor (REABA)**, an IT-based solution designed to address the user-side barriers associated with renewable-energy adoption. The system uses a simple questionnaire to understand the concerns of potential users and identifies their major adoption barriers. Based on the responses, it provides personalized information and guidance, including possible financial-support resources, basic technical information and a step-by-step adoption roadmap.

The proposed solution is different from a renewable-energy site assessment system because it does not primarily determine whether a particular physical location is suitable for renewable-energy installation. Instead, it focuses on understanding **why a user is hesitant to adopt renewable energy and what information or support could help overcome that hesitation**. The system can be implemented as a web application using HTML, CSS, JavaScript and Python with a rule-based advisory mechanism.

REABA can contribute to Sustainable Development Goal 7 by improving awareness and supporting informed consideration of clean-energy options. By addressing information and decision-making barriers, the proposed system has the potential to encourage greater renewable-energy adoption and contribute indirectly to environmental sustainability.

2. Introduction

Energy plays an important role in modern society. Households, industries, educational institutions, businesses and transportation systems depend on energy for their daily activities. For many years, a large portion of energy requirements has been met through conventional sources such as coal, petroleum and natural gas. Although these sources have supported economic development, their continued use contributes to air pollution, greenhouse gas emissions and climate-related environmental problems.

Renewable energy provides an important alternative to conventional energy sources. Renewable resources include sunlight, wind, water and biomass, which are naturally replenished. Technologies such as solar photovoltaic systems, solar water heaters and wind-energy systems can help reduce dependence on fossil fuels and support cleaner energy generation.

Despite the increasing availability of renewable-energy technologies, their adoption can still be limited. A person may be interested in renewable energy but may not know which technology is appropriate, how much it may cost, what maintenance is required, or whether financial support is available. Some users may also have concerns about reliability or may find the adoption process complicated.

Therefore, low adoption of renewable energy is not only a technological issue. It is also an **information and decision-making problem**. Providing renewable-energy technology alone may not be sufficient if potential users do not have enough information or confidence to make an adoption decision.

The proposed **Renewable Energy Adoption Barrier Advisor (REABA)** addresses this issue from the user's perspective. Instead of concentrating on the physical characteristics of a site, the system attempts to understand the user's concerns and provide targeted guidance.

The system can be accessed through a website where users answer a short questionnaire. Their responses are processed using a barrier-analysis mechanism. The system then identifies the most significant barriers and provides appropriate advisory information. For example, a user who is primarily concerned about cost can receive financial-planning information and guidance about applicable support schemes, while a user with limited awareness can receive basic educational information about renewable energy.

The proposed approach aims to make renewable-energy information more accessible and reduce uncertainty for potential adopters.

3. Problem Statement

Although renewable-energy technologies are becoming increasingly available, many potential users are still hesitant to adopt them. This creates a gap between the availability of clean-energy technologies and their actual adoption by households and other small users.

Several factors can contribute to this problem.

- **Financial concerns** are one important barrier. Potential users may consider the initial investment required for renewable-energy systems to be too high. They may also be uncertain about long-term savings and the time required to recover the initial investment.
- **Lack of awareness** can also prevent adoption. Users may not understand how renewable-energy technologies work, what options are available or how these technologies can be used in their particular situation.
- **lack of information about government schemes, subsidies and financial support.** Even when assistance is available, potential users may not know where to find reliable information or whether a particular scheme applies to them.
- **Maintenance concerns** may also discourage adoption. Users who have never used renewable-energy technology may assume that the systems require complicated maintenance or frequent servicing.
- **Reliability concerns** can create additional hesitation. Since some renewable-energy sources depend on natural conditions, users may be uncertain about whether renewable energy can meet their energy requirements consistently.

Finally, the **complexity of the adoption process** can discourage first-time users. Understanding technology choices, costs, financial assistance, installation procedures and maintenance can be difficult without accessible guidance.

The problem can therefore be stated as:

“Potential users face financial, informational and practical barriers that discourage the adoption of renewable energy. There is a need for an accessible IT-based advisory system that can identify these barriers and provide personalized information and guidance to help users move towards informed renewable-energy adoption.”

The proposed system focuses specifically on this user-side problem.

4. Sustainability Analysis

4.1 Relevant Sustainable Development Goals

The proposed Renewable Energy Adoption Barrier Advisor is primarily aligned with **Sustainable Development Goal 7 (SDG 7) – Affordable and Clean Energy**. SDG 7 promotes access to **affordable, reliable, sustainable and modern energy**. REABA supports this goal by helping users understand renewable-energy options and overcome barriers to adoption.

The system can contribute by:

- Increasing awareness of renewable-energy technologies.
- Providing accessible information to potential users.
- Helping users understand common adoption concerns.
- Directing users towards relevant financial-support information.
- Reducing uncertainty about the renewable-energy adoption process.
- Encouraging informed consideration of clean-energy alternatives.

The solution also supports **SDG 13 – Climate Action**, as increased renewable-energy adoption can help reduce dependence on fossil fuels and greenhouse-gas emissions. REABA contributes indirectly by encouraging informed adoption decisions. It also relates to **SDG 12 – Responsible Consumption and Production** by promoting awareness of cleaner energy choices and more responsible energy consumption.

4.2 ISO 14001

ISO 14001 is an international standard that provides requirements and guidance for an Environmental Management System (EMS), helping organizations manage and improve their environmental performance. REABA is a software solution and not an EMS, but its environmental-management principles are relevant to the project's sustainability objectives.

The system promotes environmental responsibility by increasing awareness of cleaner energy alternatives and encouraging users to consider the impacts of conventional energy use. If implemented by an organization, indicators such as user engagement, progress towards renewable-energy adoption and the effectiveness of awareness activities could be monitored. Thus, REABA supports the principles of environmental awareness, monitoring and continuous improvement.

4.3 Life Cycle Assessment (LCA)

Life Cycle Assessment evaluates the environmental impacts associated with a product, process or service throughout its life cycle. For renewable-energy technologies, this can include manufacturing, transportation, installation, operation and end-of-life stages.

A complete LCA is outside the scope of the proposed website because REABA is primarily an advisory software system rather than a renewable-energy product.

However, LCA is relevant when discussing the environmental benefits of renewable-energy adoption. Renewable-energy systems should not be considered completely impact-free because their manufacturing and disposal also require resources and energy.

In future versions, REABA could provide users with simplified life-cycle information about different renewable-energy technologies. This could help users understand both the benefits and environmental considerations associated with their choices.

4.4 Environmental Impact Assessment (EIA)

Environmental Impact Assessment is generally used to identify and evaluate the potential environmental effects of proposed development projects.

A formal EIA is **not directly applicable to the proposed REABA website**, because the project is a software-based advisory system and does not involve construction or direct modification of the physical environment.

However, EIA can become relevant if the system is later expanded to support larger renewable-energy projects. In such cases, information about environmental impacts, land use, biodiversity and surrounding communities could be incorporated into the advisory process.

Therefore, LCA and EIA are considered relevant concepts but are not performed as formal assessments within the proposed prototype.

5. Proposed Engineering/IT Solution

5.1 Overview of the Proposed Solution

The proposed solution is the **Renewable Energy Adoption Barrier Advisor (REABA)**, a web-based system that helps users understand and overcome barriers to renewable-energy adoption.

Users complete a simple questionnaire about their knowledge, concerns and expectations. Based on their responses, REABA identifies the main barriers, such as cost, lack of awareness, maintenance, reliability or difficulty understanding the adoption process.

The system then provides **personalized advice**, including information about renewable-energy options, financial support and the steps involved in adoption. The main aim is to help users understand their concerns and make an informed decision about renewable energy.

5.2 Main Features

The main features of REABA are:

- **User Questionnaire:** Collects information about the user's concerns and awareness.
- **Barrier Analysis:** Identifies the major barriers based on questionnaire responses.
- **Personalized Advisory:** Provides guidance according to the user's main concern.
- **Financial Support Information:** Provides information about relevant subsidies, schemes and financing options.
- **Adoption Roadmap:** Shows users the general steps they can follow towards renewable-energy adoption.

5.3 Methodology and Working

The system follows a simple questionnaire-based process. First, the user accesses the website and completes the questionnaire. The responses are then processed and assigned to different barrier categories.

The system calculates the scores for each category and identifies the major barrier. Based on the result, relevant advice and information are displayed to the user.

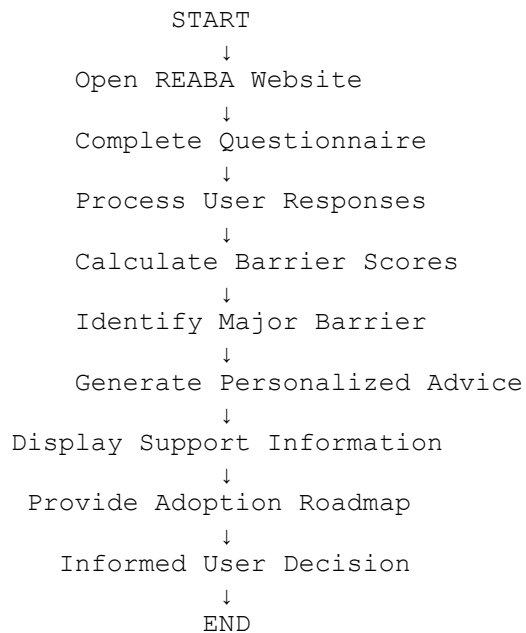
For example:

- If **financial concerns** are highest → provide information about costs and financial support.
- If **lack of awareness** is highest → provide basic renewable-energy information.
- If **maintenance concerns** are highest → provide information about maintenance requirements.
- If **reliability concerns** are highest → provide general information about system reliability.

A simple **rule-based approach** can be used to connect the questionnaire responses with suitable recommendations. This makes the system easier to develop and understand.

5.4 System Flowchart

The overall working of REABA can be represented as:



The flowchart shows how the system collects user information, analyses the responses and provides suitable guidance. The final adoption decision remains with the user.

5.5 Technologies Used

The proposed website can be developed using commonly available technologies:

- **HTML, CSS and JavaScript** – for the front end and user interface.
- **Python and Flask** – for backend processing.
- **MySQL or Firebase** – for storing relevant information.
- **Rule-based logic** – for analysing responses and generating recommendations.

The system does not require specialised hardware and can be developed and tested using a normal computer.

6. Feasibility Analysis

The feasibility of REABA can be considered from technical, economic, operational and environmental perspectives.

6.1 Technical Feasibility

REABA is technically feasible because it uses basic web-development technologies. HTML, CSS and JavaScript can be used to create the website, while Python and Flask can handle backend processing. A database can store relevant system information.

The rule-based recommendation method is also simple to implement and test. No specialised hardware is required.

6.2 Economic Feasibility

The project can be developed at a low cost because most of the required technologies are open-source and freely available. The initial prototype does not require expensive equipment.

If developed further, costs may include website hosting, database services and maintenance.

6.3 Operational Feasibility

The system is designed to be simple and user-friendly. Users only need to answer the questionnaire and view the recommendations provided.

The information about renewable-energy options and financial support can also be updated when necessary, allowing the system to remain useful over time.

6.4 Environmental Feasibility

REABA is a software-based solution and does not directly generate renewable energy or reduce emissions. Its environmental benefit is indirect.

By increasing awareness and reducing information barriers, the system may encourage users to consider renewable-energy options. Greater adoption of renewable energy can potentially reduce dependence on fossil-fuel-based energy.

7. Expected Environmental Benefits

The expected environmental benefits of REABA are mainly indirect and depend on users eventually considering or adopting renewable energy.

- **Increased awareness:** Helps users understand renewable-energy technologies and their benefits.
- **Reduced information barriers:** Makes relevant renewable-energy information easier to access.
- **Encouragement of clean-energy adoption:** Personalized guidance may help users overcome concerns that discourage adoption.
- **Reduced fossil-fuel dependence:** Greater renewable-energy use can reduce dependence on conventional energy sources.
- **Potential reduction in emissions:** Increased use of renewable energy can contribute to lower greenhouse-gas emissions.

REABA itself does not directly produce these environmental benefits. Its role is to provide information and guidance that can support informed renewable-energy decisions.

The effectiveness of the system could be evaluated through indicators such as the number of users completing the questionnaire, users accessing support information and users progressing towards renewable-energy adoption.

8. Justification of the Proposed Solution

Low adoption of renewable energy can be caused by several user-related barriers, including high initial costs, lack of awareness, maintenance concerns, reliability concerns and uncertainty about the adoption process.

REABA is proposed to address these barriers through a simple and user-focused approach. Instead of providing only general information, the system first identifies the user's main concerns and then provides relevant guidance.

The questionnaire makes the system easy to use, while the rule-based recommendation method makes the prototype simple to develop and test. The system also brings information about renewable-energy options, financial support and the adoption process together in one platform.

The solution is technically and economically practical for a student-level project and supports the objectives of **SDG 7 – Affordable and Clean Energy**. By helping users make informed decisions, REABA can contribute indirectly to greater consideration and adoption of renewable-energy options.

9. Conclusion

The **Renewable Energy Adoption Barrier Advisor (REABA)** provides a simple IT-based approach to addressing some of the barriers that prevent users from considering renewable energy.

The system uses a questionnaire to identify the user's major concerns and provides personalized advice, financial-support information and an adoption roadmap. It can be developed using basic web technologies and a simple rule-based recommendation system.

Although REABA does not directly generate renewable energy or reduce emissions, it can contribute indirectly by increasing awareness, reducing information barriers and encouraging users to consider cleaner energy alternatives.

Overall, the proposed solution is **simple, practical and user-friendly**, while supporting the objectives of **SDG 7 – Affordable and Clean Energy**.

10. References

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